

QT 511: Introduction to Quantum Optics

Course Code	QT 511
Course Name	Introduction to Quantum Optics
L-T-P-C	3-0-0-3
Prerequisites	Engineering Mathematics (Linear Algebra, Complex algebra, basics of 2nd of ODEs and initial value problems, 2nd order PDEs and boundary value problems, Probability and Statistics, Random variables). Maxwell's equations and EM theory at the level of the core physics syllabus from AICTE model
Intended for	UG/PG/PhD
Distribution	Elective PG/Elective UG
Semester	Odd/Even
Mutual Exclusion	PH 550
Approval	47th Senate (Sep 2025)

Preamble

Students of this course learn (i) To quantise the electromagnetic field, (ii) The various experimental techniques in photonics, (iii) The various representations of states of light, (iv) Classical, semi-classical and fully quantum models of light-matter interaction, (v) Modelling decoherence through Master equation

Course Contents

- **Quantization of the electromagnetic field**
 - Number states, coherent states, squeezed states
 - Hanbury-Brown and Twiss experiments – Photon bunching, Photon anti bunching
 - Hong-Ou-Mandel interference
- **Theory of Optical coherence**
 - Young's double slit experiment and first order coherence
 - Coherence functions of arbitrary order
 - Normal ordering, symmetric ordering and anti-normal ordering of operators
 - Interferometry
- **Phase-space representations of states of light**
 - Wigner distribution
 - P-function and the notion of non-classicality with some examples of nonclassical states like squeezed states and their applications
 - Husimi Q function
- **Light-matter interaction**
 - Classical model of light-matter interaction
 - Semi-classical model of light-matter interaction-
 - Quantum light-matter interaction
 - Rabi Model
 - Jayne's-cummings model
- **Open quantum systems**
 - Fermi golden rule
 - Born-Markov Lindblad Master Equation

Course Outcomes

Students of this course learn

1. To quantise the electromagnetic field
2. The various experimental techniques in photonics
3. The various representations of states of light
4. Classical, semi-classical and fully quantum models of light-matter interaction
5. Modelling decoherence through Master equation

References

1. *Introductory Quantum Optics*, Christopher Gerry and Peter Knight, Cambridge University Press (2004)
2. *Quantum Optics*, D. F. Walls, Gerard J. Milburn, 2nd Edition, Springer (2008)
3. *Quantum Optics: An introduction*, Mark Fox, Oxford University Publishers (2006)
4. *Quantum Optics for Beginners*, Z. Ficek and M. R. Wahiddin, 1st edition, Jenny Stanford Publishing (2014)